

# Why 4 MHz?

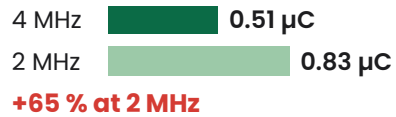
The radio sets the pace: run slow, wake fast, sleep.

## MCU charge per read (15 ms awake)

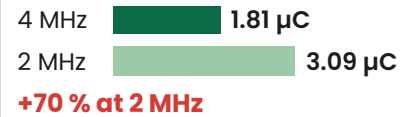


## 2 MHz vs 4 MHz, SPI 500 kHz in both

### MCU job



### ADC sample



### Card read



**12  $\mu\text{s}$**

wake-up from Stop 2, no PLL

**99.965 %**

of idle time asleep

**0.4 %**

of idle current is the MCU running

IRQ priority: **USB 5** · **reader, button, LPTIM 6** · SysTick 15